

ELEC/PHYS 450/550 Quantum Computing

1. Course Information

Instructor: Dr. Mehmet Cengiz Onbaşlı (monbasli@ku.edu.tr)

KU Credits: 3

ECTS Credits: 6

Prerequisites: none

Class Location & Meeting Times: Tue - Thu / 1:00 PM - 2:10 PM SNAB152

PS (Yes/No): Yes

DS (Yes/No): No

Lab (Yes/No): Yes (Computer Lab)

Language of Instruction: English

Office hours: after lectures, or by appointment

Teaching Assistants: Hilal Esra Yıldız (hyaldiz23@ku.edu.tr), Kerem Anar (kanar17@ku.edu.tr)

2. Course Description

Introduction to quantum mechanics for computing; quantum bits, quantum gates, and quantum circuits; quantum algorithms including Deutsch-Jozsa, Grover's search, and Shor's factoring; quantum hardware implementations; quantum error correction; quantum programming using Qiskit and Cirq; current applications and future prospects of quantum computing.

3. Course Overview

This course provides a comprehensive introduction to quantum computing, bridging theoretical quantum mechanics with practical algorithm implementation. Students will learn the fundamental principles underlying quantum computation, master key quantum algorithms, understand various hardware approaches, and gain hands-on experience programming quantum circuits.

The course is structured to accommodate students from both physics and engineering backgrounds, providing necessary quantum mechanical foundations while focusing on computational applications. Students will develop both theoretical understanding and practical programming skills through weekly coding assignments using industry-standard quantum computing frameworks.

The course has weekly reading assignments, theoretical problem sets, and programming projects. Grading is based on in-class attendance, 5 programming assignments, 1 midterm exam, and a comprehensive final project. Successful students will establish foundational knowledge of quantum computing theory, develop quantum algorithm implementation skills, understand current hardware limitations, and be prepared for advanced research or industry applications in quantum information science.

This course covers the theoretical foundations, algorithmic principles, and practical implementations of quantum computing systems. The course has six main parts:

Weeks 1-3: Quantum mechanics foundations including quantum states, superposition, entanglement, and quantum measurement theory essential for quantum computing.

Weeks 4-6: Quantum computing fundamentals including qubits, quantum gates, quantum circuits, and basic quantum algorithms with hands-on programming exercises.

Weeks 7-9: Advanced quantum algorithms including Grover's search algorithm, Shor's factoring algorithm, and quantum simulation algorithms with implementation projects.

Weeks 10-11: Quantum hardware systems including superconducting qubits, trapped ions, photonic systems, and quantum error correction methods.

Week 12: Quantum machine learning, optimization algorithms, and near-term quantum applications (NISQ era computing).

Weeks 13-14: Current research frontiers, quantum advantage demonstrations, and final project presentations on quantum computing applications.

For each section, students will engage with both theoretical foundations and practical implementations using quantum programming frameworks, preparing them for research or industry applications in quantum information science.

4. Course Learning Outcomes (CLOs):

CLO #	Upon successful completion of this course, students will be able to...
1	Understand fundamental quantum mechanical principles necessary for quantum computing including superposition, entanglement, and quantum measurement
2	Design and analyze quantum circuits using quantum gates and implement basic quantum algorithms programmatically
3	Implement and analyze major quantum algorithms including Grover's search and Shor's factoring algorithm using quantum programming frameworks
4	Evaluate different quantum hardware approaches and understand the challenges of quantum error correction and noise mitigation
5	Develop quantum computing applications and assess quantum advantage for specific computational problems

5. Assessment Methods

Method	Description	Weight (%)
Attendance	Attendance and active in-class participation to the lectures and problem sessions (70% is the minimum attendance for passing the course, for attendance $\geq 70\%$; grade = (attendance % - 70)/3)	10
Homework	5 homework assignments and their interviews Programming projects implementing quantum algorithms using Qiskit/Cirq	30
Midterm Exam	Written exam of students' knowledge and comprehension on quantum computing concepts	30
Final paper and presentation	Design and analysis paper and presentation of quantum computing (See the rubrics for the paper and the presentation)	30 (15 + 15)
	Total	100

6. Instructional Material and Learning Resources

- **Course title:** Quantum Computation and Quantum Information (10th Anniversary Edition)
Authors: Michael A. Nielsen & Isaac L. Chuang
Publisher: Cambridge University Press (2010)
ISBN: 978-1107002173

Material Status: Required

- IBM Quantum Learning: <https://quantum.cloud.ibm.com/learning/en>
- Microsoft Quantum Development Kit: <https://azure.microsoft.com/en-us/products/quantum>
- Course Page on KUHUB Learn: <https://learn.hub.ku.edu.tr/>
- KOLT Tutoring: <http://kolt.ku.edu.tr/>

7. Course Schedule

Meetings	Subject	Programming Exercise
Week 1	Introduction to quantum computing; classical vs quantum computation; overview of quantum algorithms and applications	Setting up Qiskit environment; basic Python for quantum computing
Week 2	Quantum mechanics review: states, operators, measurement; Dirac notation; quantum superposition and interference	Implementing quantum states and basic operations in Qiskit
Week 3	Quantum entanglement, Bell states, EPR paradox; quantum teleportation; no-cloning theorem	Creating and measuring entangled states; teleportation protocol
Week 4	Qubits and quantum gates; single-qubit rotations; Pauli gates; Hadamard and phase gates	Programming Assignment 1: Basic quantum circuits and single-qubit operations
Week 5	Multi-qubit systems; controlled gates; quantum circuits; reversible computation	Building multi-qubit circuits; implementing controlled operations
Week 6	Deutsch; Deutsch-Jozsa; Bernstein-Vazirani algorithms; quantum parallelism	Implementing and testing early quantum algorithms
Week 7	Grover's search algorithm: amplitude amplification, geometric interpretation, complexity analysis	Programming Assignment 2: Implementing Grover's algorithm for database search
Week 8	Quantum Fourier Transform; period finding; Shor's factoring algorithm theory	Implementing QFT and understanding its applications
Week 9	Shor's algorithm implementation; RSA cryptography and quantum cryptanalysis	Programming Assignment 3: Implementing Shor's algorithm components
Week 10	Quantum hardware: superconducting qubits, trapped ions, photonic systems; quantum error correction basics	Exploring hardware-specific constraints in quantum programming
Week 11	Variational Quantum Eigensolver (VQE); Quantum Approximate Optimization Algorithm (QAOA); quantum annealers	Programming Assignment 4: NISQ-era algorithms and applications and quantum annealer applications
Week 12	Quantum machine learning and tensor networks	Programming Assignment 5: QML demonstration and tensor network benchmarks
Week 13	Current quantum computing applications; quantum advantage and supremacy	Final project development and testing

Week 14	Final project presentations; quantum computing career paths and research opportunities	Project presentations and interviews
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8. Student Code of Conduct and Academic Grievance Procedure

Student Code of Conduct

Statement on Academic Honesty with Emphasis on Plagiarism

Academic Grievance Procedure

9. Course Policies

Students will be expected to state and sign the following Honor Code for all the assignments, exams, and projects. "I hereby certify that I have completed this quiz/assignment on my own without any help from anyone else. I have not used, accessed, received, or distributed any information from/to any other unauthorized source intaking this exam. The effort in the exam belongs only to me."

10. Other

Assessment Methods:

This course is a 3-credit required area course that consists of quantum computing lectures, problem sessions, midterm, and a final project (paper and presentation). Students are required to attend all the lectures and problem sessions for midterm and final project preparation and a better learning experience.

Final Grading:

- A passing final letter grade requires passing grades (D or above) from the course.
- A passing grade must exceed a minimum of 50% weighted average grade from the course:

$$\text{Grade} = 10\% * \text{attendance} + 30\% * \text{Programming assignments} + 30\% * \text{midterm} + 30\% * \text{final project}$$
- Passing grade = $\text{Grade} > 50\%$ and $\text{attendance} \geq 70$

Video recording policy

- Video recordings are released to the students 1 week before the midterm and the end of the semester (for midterm and final exam reviews).
- The video recordings are released at other times to the students who have a doctor's report or have a valid excuse granted by the dean of students when they are representing the country in the national sports tournaments and cannot attend the lectures for extended periods of time.

Make-up Policy

- Make-up exams are given only after a valid reason (health report, etc.) is presented.

No-cell phone or laptop policy

- During the lectures and problem sessions, students are not allowed to open their phones or laptops. Opening phones/laptops causes no attendance for that day of the lecture/problem session.
- Tablets are allowed for notetaking purposes and under "do not disturb" mode (no notifications during the sessions).